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I am a motivated and detail-oriented individual with a passion for learning and problem-solving. Mostly skilled in data science and analytics, this is my first web app. I hope it's good.

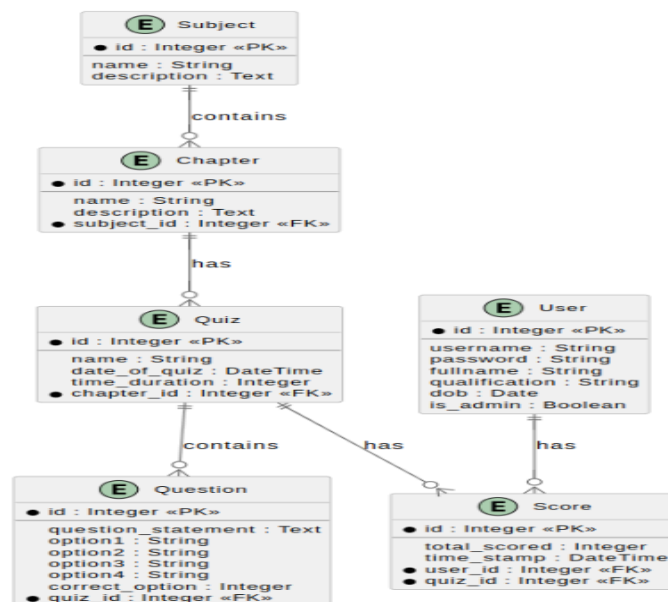
Description

This project involves building a Quiz Master Web Application where students can take quizzes, track their progress, while admins can manage quizzes, subjects. The system should be user-friendly, efficiently handle quiz data.

Technologies used

- 1. Backend - Flask:** A lightweight Python framework for backend development.
Flask-SQLAlchemy: Handles database management and object-relational mapping (ORM).
Flask-Login: Manages user authentication and sessions. **Flask-WTF:** Provides form validation and handling. **python-dotenv:** Manages environment variables and secrets.
- 2. Frontend - HTML/CSS:** Defines the structure and styling of web pages.
- 3. Database - SQLite:** Used as the database during development.

DB Schema Design



Normalization: The structure ensures efficient data storage by reducing redundancy.

Relational Integrity: Foreign keys maintain relationships between users, quizzes, and attempts.

Security: Passwords are stored as hashes for protection.

Performance: Indexing primary keys allows quick retrieval of data.

API Design

I have created APIs for user authentication, quiz management. These include endpoints for user login, quiz attempts, leaderboard access, and admin controls for managing subjects, chapters, quizzes, and questions. The API was implemented using **Flask** as the backend framework, with **Flask-RESTful** for structuring endpoints and **Flask-SQLAlchemy** for database interactions. Authentication is handled using **Flask-Login**, and request validation is managed through **Flask-WTF**.

Architecture and Features

The project follows an MVC (Model-View-Controller) structure for better code management and scalability. The **controllers** are located in the `controllers/` directory, handling API requests and logic for users, quizzes, and admin functionalities. The **models** are stored in the `models/` directory, defining database schemas with Flask-SQLAlchemy. The **templates** for the frontend are in the `templates/` folder, consisting of HTML files styled with CSS. The main Flask application file (`run.py`) initializes the app and connects various components..

The project includes core functionalities like **user authentication (login/logout), quiz management, and result tracking**. Admins can manage **subjects, chapters, quizzes, and questions** through a structured dashboard. Students can **attempt quizzes, view progress, and check the leaderboard**. These features are implemented using RESTful APIs with Flask, handling requests via JSON and integrating with an SQLite database.

Default features include quiz attempts, progress tracking, and admin quiz management. Security measures such as **password hashing and session management** are implemented to protect user data.

Video

https://drive.google.com/file/d/1q4k1aqd8mJ6n9VyTVNSSPxcr09UBp64L/view?usp=drive_link